

Introduction to SQL, MySQL & Database Fundamentals

Lecture 1 · MySQL (SQL/PL-SQL) Course

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BCA · 50–60 MINUTES



Ice Breaker 🤔

Think about the apps you used today:



Instagram



WhatsApp



Amazon



Google Pay



YouTube

"Where is all this information stored?"



Why Learn SQL?

Data Drives Technology

Every modern application — from social media to banking — is powered by data at its core.

Universal Language

SQL is the globally recognized standard language for creating, managing, and querying databases.

Developer Essential

Every software developer — frontend, backend, or full-stack — works with databases daily.

Career Multiplier

SQL skills appear in over 70% of tech job listings, making it one of the most in-demand skills.

Career Opportunities with SQL

SQL is the gateway to a wide range of high-demand tech careers:



SQL Developer

Design and optimize database queries and stored procedures.



Database Administrator

Manage performance, security, and availability of databases.



Data Analyst / BI Developer

Transform raw data into actionable business insights.



Full Stack / Backend Developer

Build server-side logic and APIs connected to databases.

Future Scope of SQL

SQL is not just for traditional databases. It powers the most transformative technologies of the 21st century. Mastering SQL today means being ready for tomorrow's innovations.

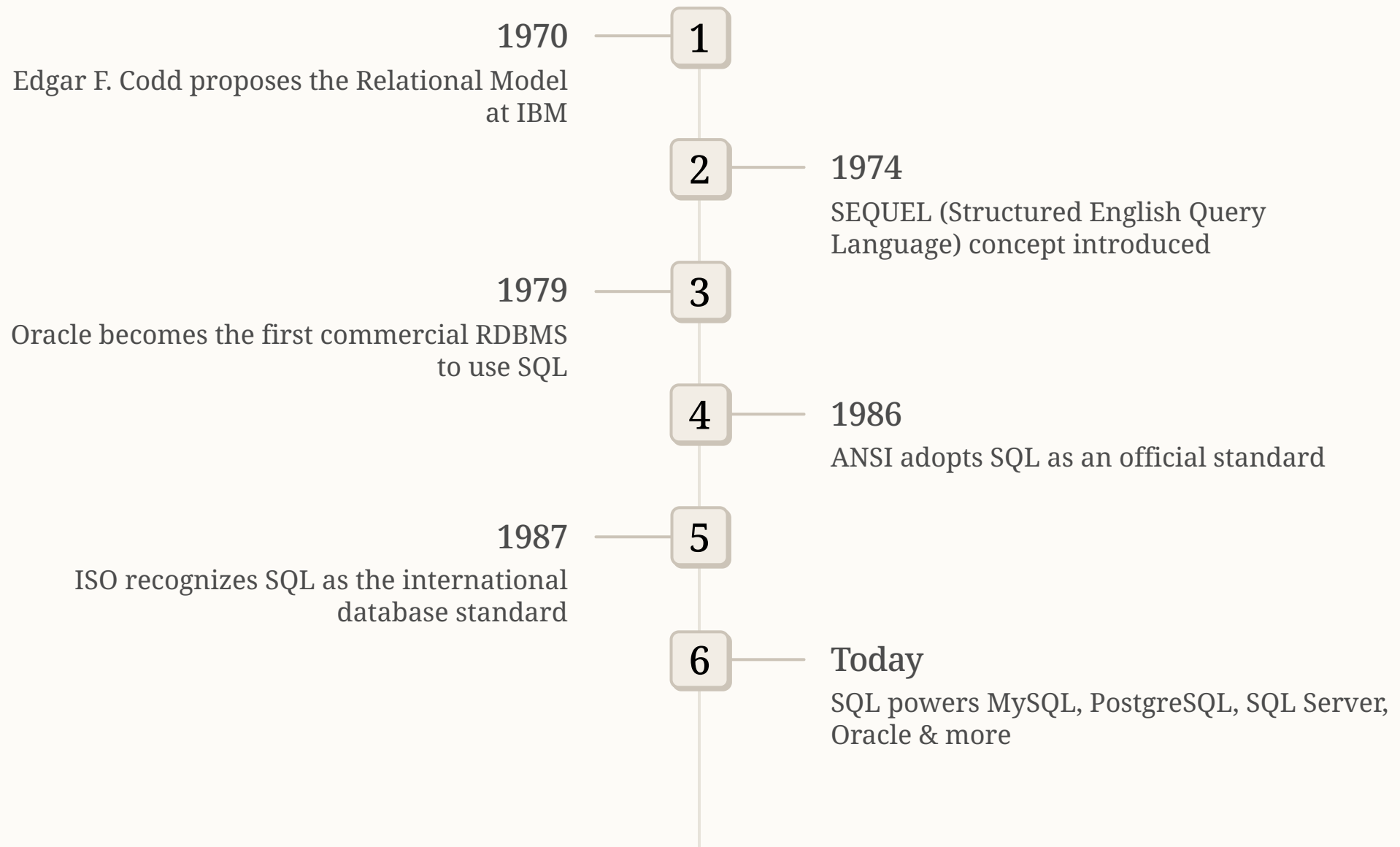


SQL skills are listed in **data science, AI/ML, and cloud roles** as a core requirement worldwide.



History of SQL

From a research paper to the world's most used data language — SQL's journey spans five decades.





What is Data?

Data is **raw, unprocessed facts and figures** — numbers, text, images, dates — that carry meaning when organized.

Student Record

Name, Roll No., Marks, DOB

Online Order

Product, Price, Quantity, Date

Patient Info

Name, Age, Blood Group, Reports

Bank Transaction

Account No., Amount, Date, Status



What is a Database?

A **database** is an organized collection of related data stored electronically so it can be easily accessed, managed, and updated.

→ Structured Storage

Data is stored in tables with rows and columns — like a well-organized digital spreadsheet.

→ Real-World Examples

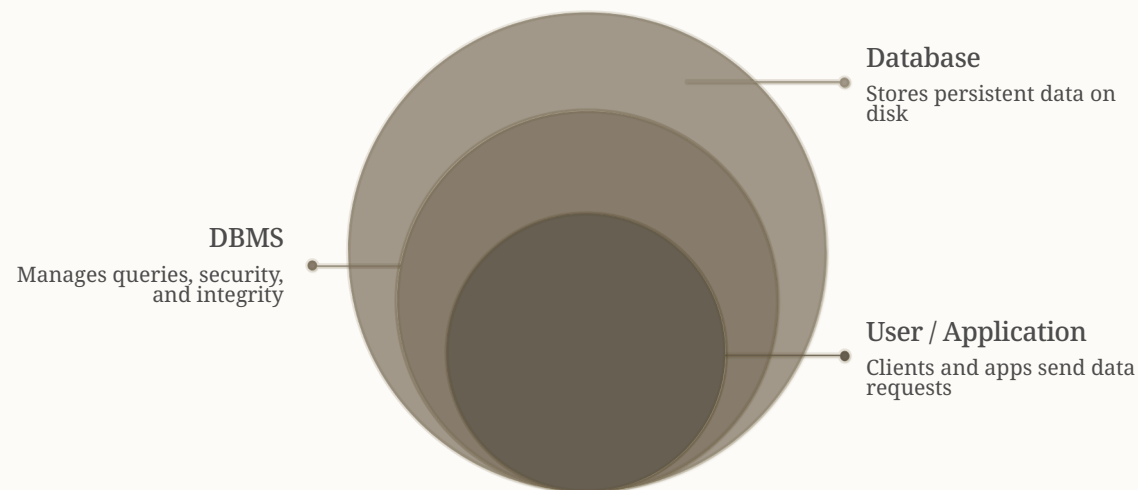
Library catalog, hospital records, bank accounts, e-commerce product listings.

→ Why It Matters

Eliminates data redundancy, ensures accuracy, and enables fast retrieval.

What is DBMS?

A **Database Management System (DBMS)** is software that acts as an interface between users and the database — allowing you to store, retrieve, and manage data efficiently.



Data Security

Controls who can access or modify data.



Data Integrity

Ensures consistency and accuracy at all times.



Easy Retrieval

Quickly fetch records using queries.

What is RDBMS?

A **Relational DBMS (RDBMS)** organizes data into **tables** that are linked by relationships. Each table has rows (records) and columns (fields).

StudentID	Name	Course	Marks
101	Priya Sharma	BCA	88
102	Rahul Verma	BCA	75
103	Sneha Patel	BCA	92

Examples of RDBMS: **MySQL, Oracle, MS SQL Server, PostgreSQL**

The Conceptual Ladder

These six concepts build upon each other — mastering them in order is the key to understanding how modern applications work.



Data

Raw facts: numbers, names, dates



Database

Organized collection of related data



DBMS

Software to manage the database



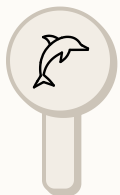
RDBMS

Relational tables with relationships



SQL

Language to interact with RDBMS



MySQL

Open-source RDBMS that uses SQL

What is SQL?

Structured Query Language (SQL) is the standard language used to communicate with relational databases. It allows you to create, read, update, and delete data.

S — Structured

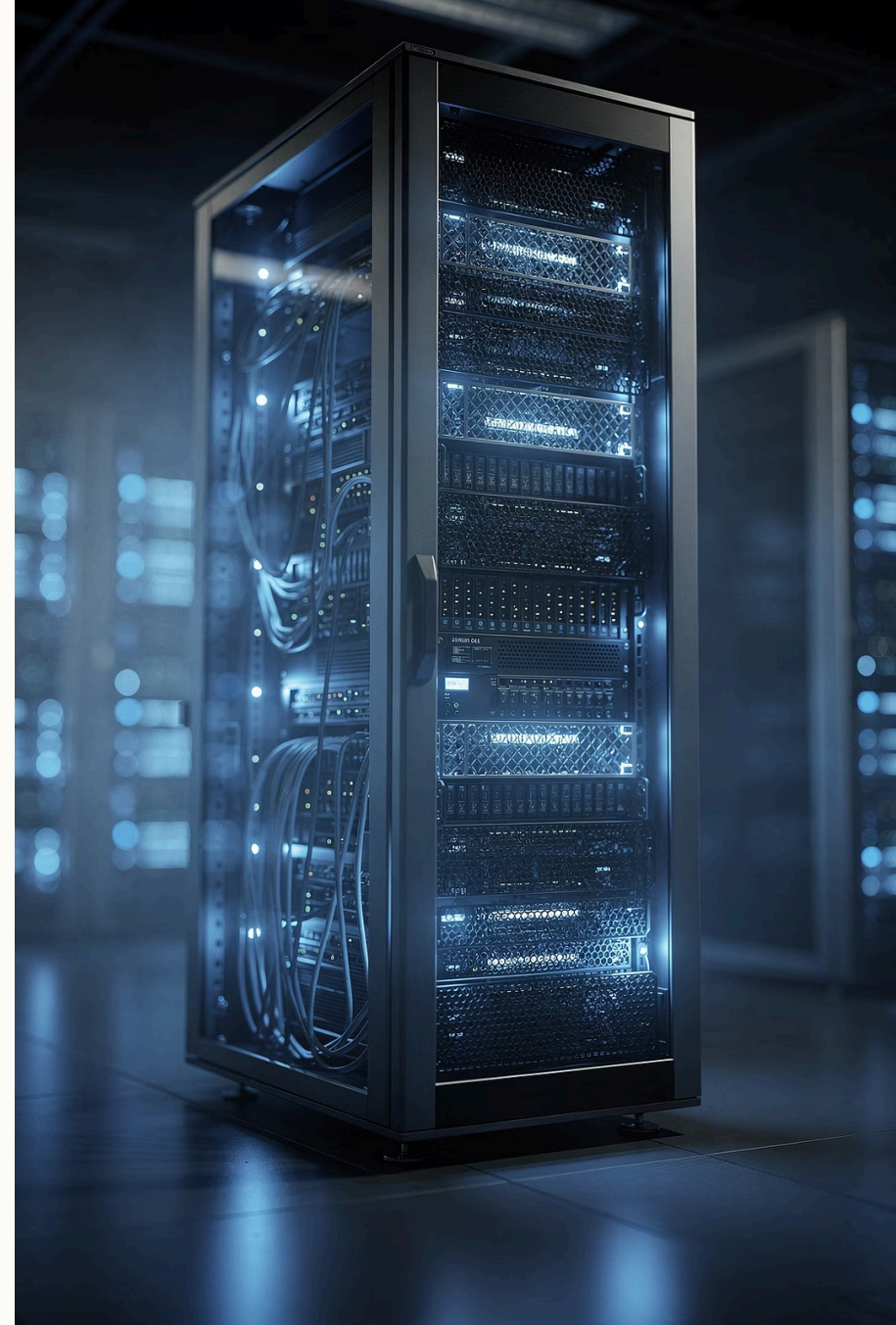
Data is organized in defined tables and schemas.

Q — Query

Used to ask the database for specific information.

L — Language

A set of commands understood by all major RDBMS platforms.





What is MySQL?

MySQL is the world's most popular **open-source Relational Database Management System**, developed by Oracle Corporation. It uses SQL as its query language.

1

Open Source

Free to download and use for most applications.

2

Fast & Reliable

Handles millions of queries per second efficiently.

3

Cross-Platform

Runs on Windows, Linux, and macOS seamlessly.

SQL vs MySQL vs SQL*Plus

Three terms that beginners often confuse — here's a clear comparison:

Feature	SQL	MySQL	SQL*Plus
Type	Language	RDBMS Software	Oracle CLI Tool
Created By	ANSI/ISO	Oracle Corp.	Oracle Corp.
Purpose	Query databases	Manage DB server	Run SQL on Oracle
Cost	Standard	Open Source	Part of Oracle DB
Used With	Any RDBMS	SQL language	Oracle RDBMS only

Categories of SQL Commands

SQL commands are grouped by their purpose. Each category controls a different aspect of the database:



DDL

Data Definition Language

Define & modify database structure

CREATE, ALTER, DROP



DML

Data Manipulation Language

Add, update, delete records

INSERT, UPDATE, DELETE



DQL

Data Query Language

Retrieve data from tables

SELECT



DCL

Data Control Language

Manage permissions & access

GRANT, REVOKE



TCL

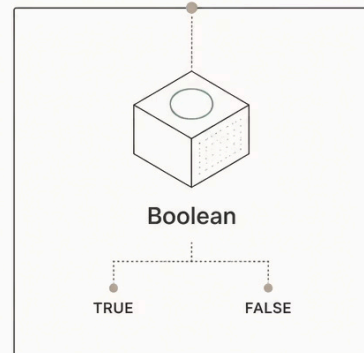
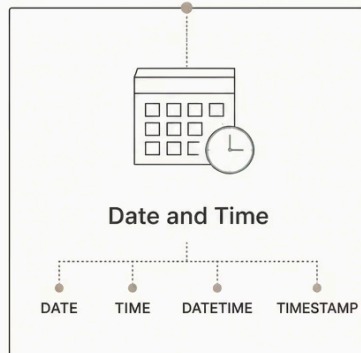
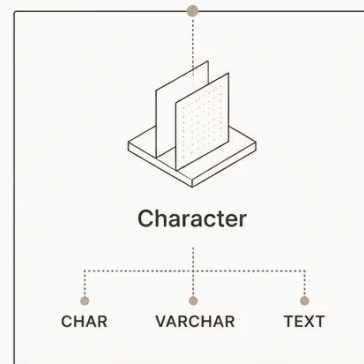
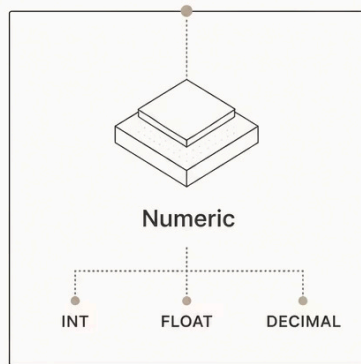
Transaction Control Language

Manage database transactions

COMMIT, ROLLBACK

Introduction to SQL Data Types

Data types tell MySQL **what kind of value** a column will store. We'll explore these in detail in future lectures.



Numeric

Whole numbers, decimals, and floating-point values. e.g., marks, salary, age.

Character

Text strings — names, addresses, emails. Fixed or variable length.

Date & Time

Storing dates, times, and timestamps for records and transactions.

Boolean

Logical TRUE / FALSE values for conditions and flags.



Detailed syntax and examples will be covered in **Unit I** of this course.

Course Roadmap

Your complete journey through SQL and PL/SQL — from fundamentals to real-world projects:

1

Foundations

SQL Intro, Data Types, Operators

2

Core SQL

DDL, DML, DQL Commands & Tables

3

Advanced Queries

Functions, Joins, Subqueries

4

DB Objects

Views, Indexes, Transactions

5

PL/SQL

Procedures, Functions, Triggers

6

Mini Project

Real-world database application

Software You Will Need

Set up your environment before the next class. All tools below are free to download.



MySQL Community Server

The core database engine. Download from mysql.com — choose version 8.x.



MySQL Workbench

Official GUI for MySQL. Write and execute SQL queries visually.



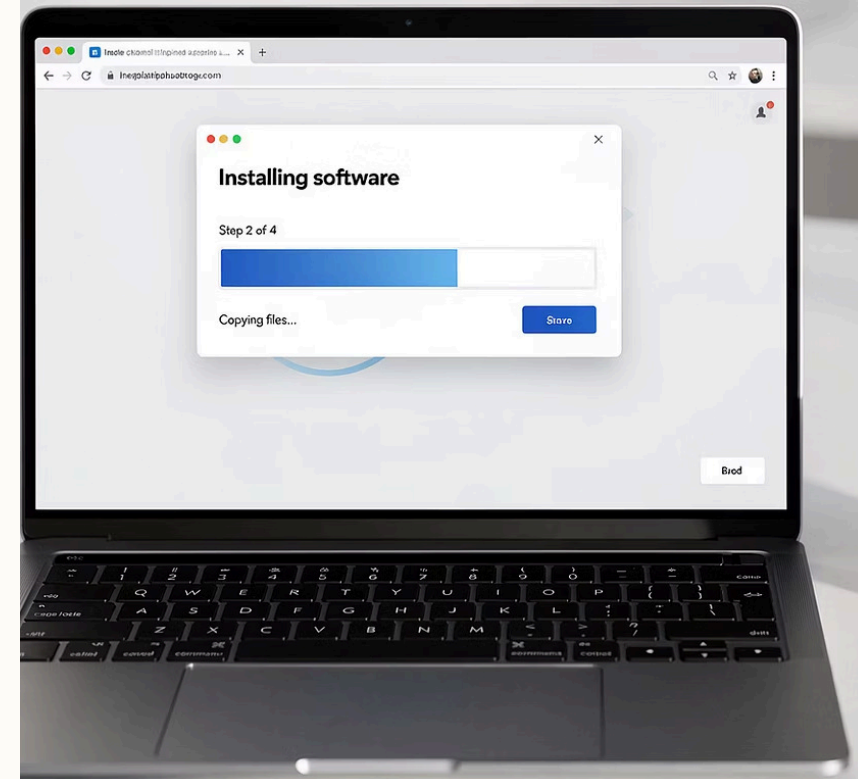
SQL*Plus

Oracle's command-line tool. Covered conceptually as part of the BCA syllabus.



VS Code (Optional)

A powerful code editor with SQL extensions for writing scripts.



Quiz & Assignment

Quick Quiz

1. What does SQL stand for?
2. Who proposed the Relational Model in 1970?
3. Which category does the CREATE command belong to?
4. Name two examples of RDBMS software.
5. What is the difference between a row and a column in a table?

Assignment

Research and write a **one-page comparison** on the following:

- What is SQL? What is MySQL?
- What is Oracle Database?
- What is SQL*Plus and when is it used?
- Key differences between SQL and MySQL

 Submit handwritten or typed — due **next class**.

Key Takeaways & Next Lecture

Key Takeaways

- SQL (Structured Query Language) is the standard language for managing relational databases
- A Database is an organized collection of structured data stored electronically
- DBMS acts as an interface between users and the database for efficient data management
- RDBMS organizes data into interrelated tables using rows and columns
- SQL commands are categorized into DDL, DML, DCL, TCL, and DQL

Assignment

- Research and list 5 real-world applications that use SQL databases in their backend
- Write the difference between DBMS and RDBMS in your own words with one real-life example each
- Install MySQL Community Server and MySQL Workbench on your system and take a screenshot of the successful installation

Next Lecture

SQL Data Types & Constraints

We will explore numeric, string, and date data types — and how constraints like **PRIMARY KEY**, **NOT NULL**, and **UNIQUE** enforce data integrity.

Discussion & Doubts

Questions, Discussions & Doubts

Learning never ends. Keep practicing, stay curious, and prepare for the next lecture.